Kurvst Arrowhead Engine

SOURCE31 classifies this as a historical design and implementation investigation, preserved from frozen `a517ca563d78d161cff17fe60e661f80529c9a83`. Status, "today" and migration statements below describe their original staged context, not a fresh certification of every API example. Current native graph construction uses Node/Edge/HalfEdge Spec values and `symbolica.community.graph`; current rendering owns native SVG and static fallbacks. Source38 geometry and notice changes now have a separate execution record. The original design and Tiptoe comparisons below remain historical provenance, not current geometry or notice requirements. Reported measurements retain their stated old baseline. The current Kurvst manual and registered drawing architecture own usage guidance.

> **Status: Accepted design, recorded 2026-10-04; P0–P7 implemented.** Linnest draws with stock CeTZ 0.5.2 and the shared Kurvst mark engine; no CeTZ is vendored. Performance and image acceptance remain open decisions.

Decision and delivery boundary

Move numeric arrowhead geometry, label-candidate footprints, hit-area placement, and collision sampling into Kurvst. One Rust geometry engine serves native Python SVG and Typst Wasm; stock CeTZ paints finished geometry. Kurvst owns curve and mark geometry; Linnest owns placement decisions, label search, and link semantics; CeTZ owns painting and its existing Boolean operations. Eventually delete the vendored CeTZ tree: 47 files, five core patches, and its pinned rebuild recipe. Do not port LGPL CeTZ patches into the new MIT engine.

The prerequisite Kurvst consolidation lands with this note: the shared curve API, stroke outlines, arcs, pattern splitting, and removal of the `optimize` and close-mode options. Delivery order is: consolidate, capture the P0 corpus baseline against the consolidated base, then implement the phases below.

Mark catalogue and data contract

`kurvst.mark` constructors produce data only. The Python `Mark(name, ...)` of `linnet_py` and `symbolica.community.graph` mirrors the same names, fields, and defaults and serializes the same dictionary; there is no Python `kurvst` module. Names, parameters, and defaults follow tiptoe 0.4.0 (MIT), with attribution retained in the fresh implementation.

The catalogue has thirteen shapes plus `combine`. Percent lengths below are ratios of line stroke thickness, not path length. Mixed dimensions add fixed length and stroke-relative length.

Mark	Accepted size / shape defaults	Placement / prior use
`triangle`	Stealth geometry with inset 0%	Native flow today
`straight`	Length 3pt + 450%	Native momentum today
`stealth`	Length 3pt + 450%, inset 40%	CeTZ today
`round`	Stealth with round joins	New
`tikz`	Width 3pt + 450%	New
`barb`	Width 3pt + 450%, arc 180deg	New
`hooks`	Width 3pt + 450%, arc 180deg	New
`bar`	Width 2.4pt + 360%	Centred
`bracket`	Width 2.4pt + 360%	End aligned

Mark	Accepted size / shape defaults	Placement / prior use
`circle`	Length 400%	Centred
`square`	Length 400%	Centred
`diamond`	Length 565.69%	Centred
`rays`	n = 4, length 280%	Centred
`combine`	Marks and gaps in sequence	One composite

Parameters include `length`, `width`, `inset`, `fill`, `stroke`, `rev`, `align`, `arc`, `n`, and `phase`. Automatic width follows the per-shape ratios. Lengths may be fixed, stroke-width ratios, or mixed, such as `3pt + 450%`. Fill defaults to the line stroke paint; sizing uses the line stroke thickness. The tiptoe end distance defines the shaft stop, including the stealth inset notch. Arrow tips land on the path end. Bar, circle, square, diamond, and rays are centred unless `align: "end"`.

`combine` is one composite with aggregate contacts, end distance, and footprint. Apply the end fit once to the composite, not independently to its children.

Carrier, placement, and curved-end fits

Preserve `mark-position`, `mark-orientation`, `mark-direction`, and `mark-shift`, including centre placement, arc-length ratios, signed shifts, inward endpoint clamping, and label footprints. Stations and shifts use the derived visible carrier **after offsets and node outsets, before arrow shortening or bending**. Footprints and collision geometry instead consume the painted shaft. Kurvst arc length is the chosen source of truth: accept its tiny numerical drift while keeping Python and Typst geometry identical.

Fit	Accepted rule
`chord` (default)	Put tip and back on the original carrier; trim the shaft at the back without changing the carrier. This preserves the current arrow behavior.
`bend`	Follow the end tangent and retract the endpoint by mark end distance times `shorten`, retaining the control points as tiptoe does. Default `shorten` is 100%; 0% is unbent.

Fits apply at ends. Interior marks straddle their station on a chord. The painted shaft owns footprint and collision geometry; path patterns follow the bent base when `bend` is selected, including phase behavior.

The curved-end comparison has no image asset in this workspace. Its intended reference uses the same cubic and a 22 × 16pt triangle: `bend` with `shorten: 0%` leaves the line exiting the side; `bend` at 100% reshapes the endpoint to the head centre; `chord` turns the head to the original carrier while leaving the curve unchanged. Large heads exaggerate the differences. P0 will generate real references; no missing `curved-ends.png` is linked here.

Execution boundary: two bounded batches

Each drawing uses **two bounded batches**, both sharing templates and the engine:

1. Prepare candidate geometry and footprints for Linnest's label search.
2. Paint the selected placements using finished geometry.

This is not a one-call design. Candidate preparation avoids repeated Typst geometry evaluation; numeric hitboxes and prepared content drawables avoid per-target painting work. Link target selection and semantics remain Linnest responsibilities, separate from the mark catalogue.

API and migration

Mark constructors, the native Python configuration and the Typst renderer migration are implemented as sketched below.

```
#import "../../crates/kurvst/typst/lib.typ": kurvst
#let flow = kurvst.mark.triangle(length: 0.21cm, width: 0.1575cm)
#let momentum = kurvst.mark.straight(length: 0.16cm, width: 0.12cm)
#let cut = kurvst.mark.combine(kurvst.mark.bar(), 2pt, kurvst.mark.bar())
#let bent = kurvst.mark.stealth(fit: "bend", shorten: 100%)
#let source-style = (
  stroke: ink + 0.7pt,
  mark: flow,
  mark-position: "center-if-dangling",
  mark-orientation: "edge",
)
// source-style is supplied to the drawing style configuration.
```

```
from symbolica.community.graph import GraphStyleOptions, Mark, RenderSettings

flow = Mark("triangle", length="0.21cm", width="0.1575cm")
momentum = Mark("stealth", inset="25%", fit="bend")
settings = RenderSettings(
    style=GraphStyleOptions(
        edge_style={"flow-arrow": flow, "momentum-arrow": momentum}
    )
)
diagram.render(config=settings, momenta=True)
```

Native configuration gains `flow-arrow` and `momentum-arrow`. Reject CeTZ mark dictionaries with an error showing the replacement; remove the Python `Mark` CeTZ fields `scale`, `anchor`, and `shorten_to`, with useful migration errors. Switch physics `fermion-arrow-mark` and `momentum-arrow-mark` together with documentation and examples. Preserve the notebook look using explicit existing sizes: tiptoe defaults are larger (roughly 7.5pt for straight at a 1pt line, versus roughly 4.5pt today).

On the measured feynkit revision, native marks are hardcoded: momentum chevron 0.16 × 0.12cm and flow triangle 0.21 × 0.1575cm. There are no native arrow options there. Current Typst vendored-only calls are `mark.geometry`, `mark.footprints`, and `draw.content-many`; their replacements are planned, not present here.

Reported measurements and rationale

These figures were supplied with the accepted plan and were **not reproduced** for this documentation change. Context: 24 drawing-corpus diagrams, Typst 0.15.0, feynkit `78c33d71`, measured 2026-10-04; medians of three interleaved runs.

Renderer	Total	Momentum	Plain
Vendored CeTZ	26.7s	1.3s	1.0s
Vendored, three Linnest batched calls replaced	42.3s	2.4s	1.2s
Stock CeTZ 0.5.1	70.0s	5.5s	1.5s

Stock rasterized all 24 diagrams to the same pixels in that reported comparison; SVG differences were roughly 300 invisible hit rectangles. This observation is not a blanket pixel-identity requirement for the migration.

Of stock's extra 4.75s momentum time, candidate footprints account for 79%, hit areas 16%, collision lines 2%, and generic drawing 3%. Of the extra 0.6s plain time, hit areas account for 75%, collision lines 14%, and generic drawing 9%. Stock momentum used 2.7 million Typst calls versus 186 thousand; the Wasm interpreter was reported to cost roughly 20 times native execution. These observations motivate numeric preparation rather than merely replacing the painter. P0 must record host, toolchain, cache state, timing boundaries, and paired baseline comparisons before using these figures as performance gates.

Delivery phases and evidence

Each phase is its own change after the consolidated base, with corpus timings recorded. S/M/L are planning sizes, not completion indicators.

P0 — S: reproducible corpus and visual policy

Check in the 24-diagram corpus, regenerated templates, median timing script, and PNG references. Establish explicit lenient tolerances and image-diff review for arrows and labels, **not blanket pixel identity**. Require cross-renderer outline agreement and correct link targets and hit areas. Record provenance (host, toolchain, cache, timing boundaries) and paired baseline comparison. Regenerate this baseline only after consolidation.

The P0 harness, immutable input corpus, and reference images are recorded in `tests/drawing-bench/`. The captured three-round baseline is 15.010 s on M4 Pro with Typst 0.15.1, with 24/24 reference PNGs byte-identical to the supplied snapshot at 150 ppi; the consolidated sources reproduce all 24 PNGs byte-for-byte on Linux. Source, package, host, and raw timing identities accompany the images. Numerical tolerances for future arrow/label changes still require explicit review; renderer migration is not claimed.

P1 — L: shared geometry engine, no renderer migration

Implement fresh MIT Kurvst templates for all thirteen marks and combine, sizing, end distances, placement, both fits, inward clamps, contacts, painted shaft, and footprints. Add batched native/plugin APIs and data constructors. No user-visible renderer migration yet. Compare straight shapes and end distances with tiptoe; test bend/shorten including short last segments, painted-footprint coverage, shaft/back contact, zero-length paths, and near-end placement.

P1 now provides the shared native/Wasm engine and strict data-only Typst constructors for thirteen shapes and composites. Its 110 native tests and native/Wasm parity checks pass. The approved parity tolerance is `1e-12 * max(1, abs(actual), abs(expected))` for geometry coordinates and lengths only; structure, styles, indices, and other data remain exact. Candidate results remain independent, while selected results share one authoritative painted shaft per carrier.

P2 — M: native and Python

Replace hardcoded `linnest::svg::marks` with the engine, add native `flow-arrow` / `momentum-arrow` and the Python mirror. Preserve the default notebook look with explicit sizes.

The native renderer uses one drawing-wide candidate call and one selected call, with deduplicated templates and one painted shaft per carrier. Linnest's SVG tests cover the call count, every catalogue shape and composites with both fits, independent leaf paints and hit areas, and painted candidate bounds. Native configuration reads `flow-arrow` and `momentum-arrow` from the graph edge style (`GraphStyleOptions.edge_style`); linnet-py turns `Mark` values and their public dictionaries into the numeric `MarkSpec` and its leaf paints on every native path, for ordinary graphs and domain scenes alike. It checks them, and dictionaries under the Typst `mark` key, when a style is configured, so a CeTZ dictionary or an invalid mark is an error before either renderer is chosen. A domain scene whose other settings need Typst draws the same configured heads there. Named paints resolve

through Typst's own color constants, not CSS names. Painted head outlines count in the native label-overlap score, so label feedback moves labels off heads as well as shafts.

The maintainer approved the shared engine's default appearance on 2026-10-09: retain the explicit physical sizes and inset stroke geometry rather than recreating legacy protruding miters. The isolated raw-paint comparison at 150ppi changed 67/2814 pixels for the flow triangle and 74/2814 for momentum; maximum bounding-extent changes were 0.4272pt and 1.6512pt respectively. This approval does not waive corpus image review or native/Typst parity.

P3 — M: Typst painting
Replace `mark.geometry` with engine geometry painted through CeTZ; accept only Kurvst marks. Migrate physics, manual, and map-style uses. Remove one of the three vendored-only calls.

The Typst route consumes the shared candidate and selected batches. Physics templates, the map-style example, linnet-py's scene template and Python selectors use canonical mark data; CeTZ edge-mark dictionaries are rejected. All Clinnet tests, including the native/Wasm mark parity checks, pass with stock CeTZ 0.5.2 as the only package source. Named preparation anchors remain on the unshortened reference carrier; final painted heads contribute their engine outlines to conservative canvas bounds.

P4 — M: candidate footprints
Replace `mark.footprints` and the slow paint fallback with shared candidate-batch engine footprints. This addresses the main 79% momentum penalty.

Candidate templates and placement indices are collected drawing-wide before label search. Outlines determine collision boxes; retained sizing thickness cannot turn a shaft with no paint into an obstacle. This is implemented and covered by focused tests, not yet a corpus performance claim.

P5 — S: hit areas
Use numeric hitboxes and prepared content drawables instead of per-target CeTZ `content-many`. Address roughly 75% of the plain penalty and 16% of momentum's; keep link semantics separate from the catalogue.

The authored fixed-size identity-target helper passes focused stock CeTZ checks, preserving both expected SVG links. A document whose own canvas uses stock CeTZ 0.5.1 or 0.5.2 can embed Linnest edge halves, Kurvst paths and a full drawing with Kurvst heads; both produce the same SVG.

P6 — S: collision sampling
Land the collision-sampling prototype JJ change `tvxztqnq` with one plugin batch. Its reported vendored-corpus improvement was 1.8%; collision accounts for roughly 2–14% of the stock penalty. These figures are not new measurements.

The prototype revision was unavailable. The implementation instead reuses the existing Linnest `label_stroke_lines` owner, with strict numeric validation and one batched traversal of painted strokes. Twenty-three label-placement tests and a focused stock-CeTZ fixture pass. The drawing owner now gathers all painted strokes for one traversal; timing acceptance remains pending.

P7 — M: stock CeTZ and vendor removal
Target **stock CeTZ 0.5.2** (the measurements above used 0.5.1). Delete vendor, patches, and `REBUILD`; migrate behavior fixtures and Nix packaging; rerun the corpus. Generic bounds/sample speedups may optionally be upstreamed.

The patched 47-file package tree, five patches and rebuild recipe are deleted, and no CeTZ copy remains in the repository or the wheels. Nix selects stock CeTZ 0.5.2 for the documentation and the

development shell. linnet-py's Typst route reads `@preview/cetz:0.5.2` from the local package path and cache, in the Typst CLI's order, and never downloads; a missing package is reported with the directories searched. Native rendering needs no Typst package.

Validation and remaining acceptance

Measured on itphlies (AMD EPYC 9754, Linux, Typst 0.15.1) at a shared-host load average near 70. The 24-case corpus compiles with stock CeTZ 0.5.2 as the only package source. At 150 ppi, six PNGs equal P0 byte for byte and eighteen differ. Every momentum case redraws its momentum heads, and particle-flow heads are filled with the line paint instead of ink, with the engine's explicit sizes and inset stroke geometry. Label search places labels differently in some momentum cases: nonplanar K3,3 no longer overlaps two labels, the Higgs scattering figure is 8 px shorter, and in the six-leg Higgs theta figure one label now crosses a dotted edge. Image review remains required.

Paired, interleaved fresh-process compiles over five rounds compare these sources with the P0 sources and the formerly vendored, patched CeTZ 0.5.1: stock CeTZ is 1.167 times slower in the median round (range 1.164–1.224), with sums of case medians of 32.6 s against 27.7 s. Momentum cases take 1.23–1.40 times as long; particle and notebook cases 0.97–1.02 times. Performance acceptance is therefore still open; the M4 Pro ceilings above are not comparable on this host.

Acceptance and risks

On the recorded reference setup, with paired comparisons, the stock corpus must reach at most 26.7s total, 1.3s momentum, and 1.0s plain. Native rendering must not regress from the reported 60–100ms four-loop baseline. P0 must make the reference setup reproducible rather than treating these numbers as portable.

Both renderers must use the same geometry for thirteen marks plus combine. Bend must match tiptoe within the recorded tolerance; chord must match current arrows under the P0 visual policy. Cross-renderer outlines, correct links and hit areas, painted footprint coverage, and shaft contacts remain correctness requirements. Completion also requires no vendored-only calls, vendor removal, and passing stock-CeTZ tests.

- Interpreter fuel and instruction count: keep candidate loops free of trigonometry and powers; cache templates and arc-length tables.
- Label drift: review image differences with explicit lenient tolerances; tiny Kurvst arc-length drift is chosen, not hidden by separate renderer rules.
- Breaking API: migrate both arrow options, examples, and physics styles together; reject obsolete dictionaries and fields with actionable errors.
- Bent path patterns: validate phase against the painted bent base.
- Licensing: fresh MIT geometry with tiptoe attribution, not copied LGPL CeTZ patch code.

P0–P7 are implemented. The performance targets and the image review above remain open acceptance decisions.